

ClusterMind

AI-Powered Self-Healing Cluster Intelligence

Concept Note — AI Innovation Hackathon: From Learning to Impact | Track: AI for Cluster Intelligence

1. TEAM DETAILS

Team Name

Runtime Terrors

Team Members

#	Name	Role
1	MD Irfanur Islam Rahat	Team Lead — System Architecture & Full-Stack Development
2	MD Tanjimul Islam	Developer — Implementation
3	MD Shanto Ali	Software Quality Assurance (SQA)

Mentor

Md. Mezbaul Islam Zion — Software Engineer; Faculty, Department of Computer Science and Engineering, Daffodil International University.

Team Summary

Runtime Terrors is a three-member student team from the Department of Computer Science and Engineering, Daffodil International University, brought together by a shared frustration with how compute infrastructure fails: silently, expensively, and usually without warning. Our skill set spans the full delivery pipeline — system architecture and full-stack development, backend implementation, and rigorous quality assurance — and we are mentored by an industry software engineer who also teaches at DIU. Our shared motivation is simple: monitoring tools tell you something is wrong after the damage is done. We wanted to build something that acts before the damage happens, and does so without losing a single byte of active work.

2. CORE IDEA

Main Concept

ClusterMind is an AI-powered self-healing system for GPU/CPU clusters. Lightweight agents (Prometheus Node Exporter and NVIDIA DCGM) stream six live health signals — CPU usage, RAM usage, disk I/O, network traffic, GPU temperature, and GPU utilization — every few seconds into an IsolationForest anomaly-detection model. The model converts these signals into a single risk score (0–100%). When a node's risk score crosses a critical threshold, ClusterMind does not just raise an alert — it automatically checkpoints the node's active workload, migrates it to a healthy node, and restores execution exactly where it left off, all without human intervention.

Problem Addressed

GPU clusters fail silently. Industry incidents — a 2022 configuration-error cascade at a major AI research cluster, a 2023 regional GPU-zone outage with no automatic failover, and a 2024 thermal event detected only after 30–45 minutes — show that today's monitoring stacks (Grafana, Datadog) only tell teams a failure happened after it already

cost them hours of training time and thousands of dollars. This problem is faced by AI startups training models on rented or owned GPU clusters, university and research labs running shared HPC infrastructure, and cloud GPU providers managing large fleets of nodes.

Proposed Solution

ClusterMind replaces the industry's reactive loop — failure happens, an alert fires, a human investigates, a human fixes it, 45+ minutes are lost — with a predictive, autonomous loop: predict the failure 30–60 minutes early, alert through a live dashboard, and heal the system automatically with zero data loss.

- Multi-signal ML detection — IsolationForest model trained on real production cluster data, analyzing six live metrics simultaneously.
- Real-time dashboard — node health heatmap, severity levels, and cost-waste flags served through a live API and React UI.
- Checkpoint-then-migrate auto-healing — the at-risk node's workload state is snapshotted before it is isolated, then restored on a healthy node.
- Full audit trail — every detection, migration, and recovery is logged with the time and cost saved.

Unique / Innovative Aspect

Existing tools stop at monitoring and alerting; none of them close the loop by acting on the alert. ClusterMind's differentiator is the combination of early, multi-signal ML prediction with a zero-data-loss, checkpoint-based auto-healing mechanism — validated on real heterogeneous hardware (a MacBook Air M2, an NVIDIA GPU laptop, and a non-GPU laptop) rather than a simulated environment. Judges should care because this is the difference between a dashboard that reports damage and a system that prevents it.

3. FEASIBILITY & GROWTH POTENTIAL

Realistic Implementation

Yes. The core pipeline — telemetry collection, IsolationForest scoring, and workload migration — is built entirely on open-source, freely available components (Prometheus, NVIDIA DCGM, scikit-learn, Python, React) and was demonstrated end-to-end on a real three-node heterogeneous cluster within an 8-hour hackathon window. Required resources are modest: a small cluster of nodes running node exporters, a Python ML environment, and a lightweight dashboard service.

Practicality

Yes. The system requires no proprietary hardware or vendor lock-in. It runs on any cluster — bare-metal, virtualized, or Kubernetes-based — that can expose standard Prometheus-compatible metrics, making it deployable in real research labs, startups, and cloud environments with minimal setup changes.

Market Differentiation

Grafana and Datadog offer real-time monitoring and alerting but no failure prediction and no auto-healing. ClusterMind adds ML-powered prediction, automatic cost-waste detection, and — most importantly — autonomous, zero-data-loss healing, a category no current monitoring provider fully addresses.

Growth Potential

- Native Kubernetes integration for auto-scaling and scheduler-level enforcement of Safe Mode.

- Expansion of the anomaly model to additional signals (network topology, storage latency) and support for ensemble models.
- A SaaS offering for GPU cloud providers, billed on cost savings and uptime delivered.
- Multi-cluster and multi-region failover, extending today's single-cluster Safe Mode logic.

4. TECHNOLOGY STACK & AI TOOLS

Category	Details
Programming Languages	Python, JavaScript / TypeScript
Frameworks & APIs	React, FastAPI, Prometheus HTTP API, NVIDIA DCGM Exporter
AI Tools	scikit-learn (IsolationForest), custom multi-signal anomaly-scoring pipeline
Cloud & Infra Tools	Docker, Kubernetes (roadmap), Prometheus, Grafana
Security Protocols	JWT-based session auth, TLS in transit, role-based access control, checkpoint data-integrity verification

5. PROJECTED IMPACT

Target Users

AI startups training models on GPU clusters, university and research labs running shared HPC infrastructure, cloud GPU providers managing fleets of nodes, and Kubernetes-based ML platform teams.

Impact Metrics

~44 min saved per incident (45 min to <30 sec)	\$590–\$3.6K cost saved per incident	\$38,980 projected annual savings per 10-node cluster	Zero data loss during node migration
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Social & Economic Impact

Every idle or failing GPU that goes undetected wastes energy and money and slows down AI research that increasingly underpins scientific and economic progress. By catching failures 30–60 minutes before they happen and healing them automatically with no data loss, ClusterMind reduces wasted compute and energy consumption, lowers operating costs for under-resourced research labs and startups, and increases the overall reliability of the AI infrastructure that the broader AI ecosystem depends on.